

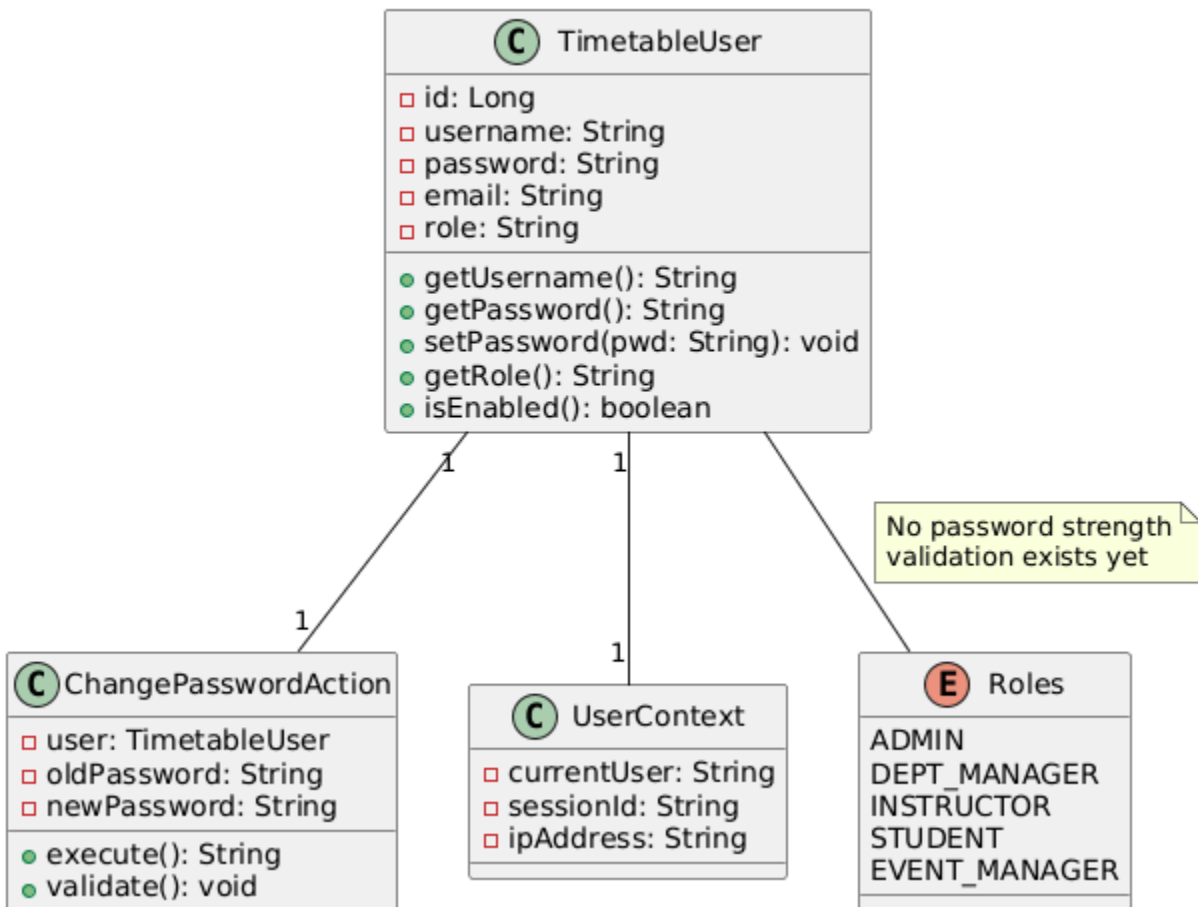
SW Maintenance & Evolution Phase 2

Team Members

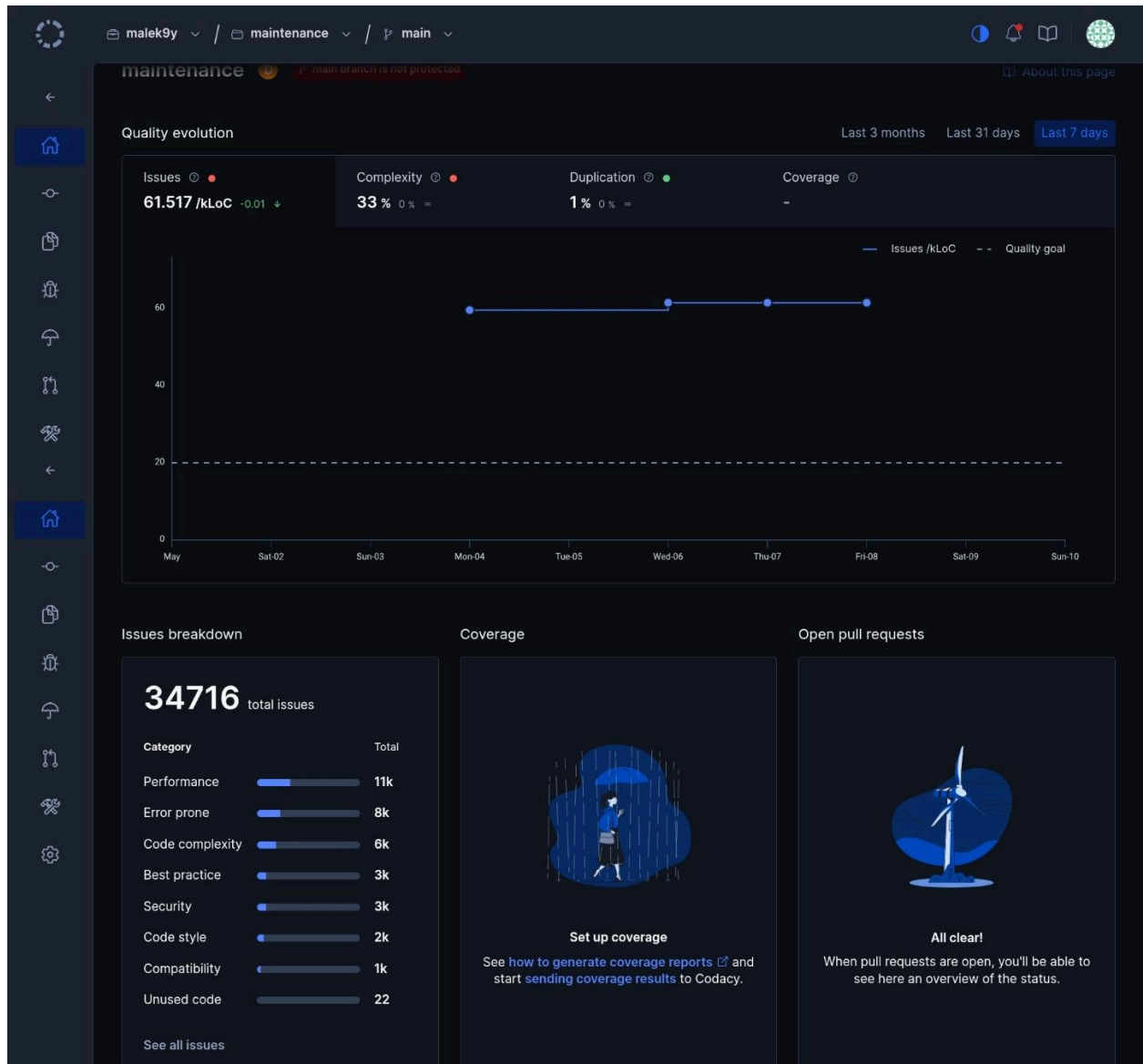
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1-Uml Diagram Before Changes

UniTime - User Management Subsystem (BEFORE Changes)



2-Impact Analysis After Changes

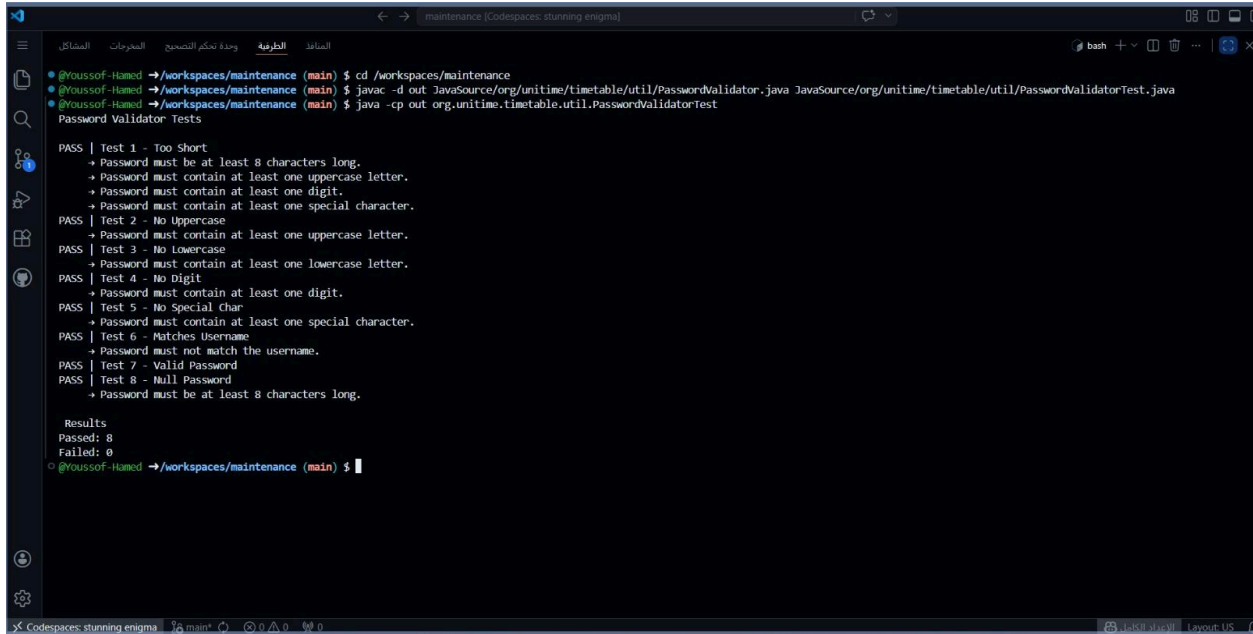


3-Verification&Validation of The Changes

⇒**Password Strength Validator**

Verification: Unit Test Results

Test No.	Test Name	Input pass	Username	Expected	ResultUser
1	Too short	abc	user	invalid	Pass
2	No uppercase	password1!	user	invalid	Pass
3	No lowercase	PASSWORD1!	user	invalid	Pass
4	No digit	Password!	user	invalid	Pass
5	No special char	Password1	user	invalid	Pass
6	Contains username	Admin123!	Admin123!	invalid	Pass
7	Valid pass	SecureP@ss1	user	valid	Pass
8	Null pass	null	user	invalid	Pass



```
maintenance (Codespaces, stunning-emoji)
@Youssof-Hamed →/workspaces/maintenance (main) $ cd /workspaces/maintenance
@Youssof-Hamed →/workspaces/maintenance (main) $ javac -d out JavaSource/org/unitime/timetable/util/PasswordValidator.java JavaSource/org/unitime/timetable/util/PasswordValidatorTest.java
@Youssof-Hamed →/workspaces/maintenance (main) $ java -cp out org.unitime.timetable.util.PasswordValidatorTest
Password Validator Tests

PASS | Test 1 - Too Short
→ Password must be at least 8 characters long.
→ Password must contain at least one uppercase letter.
→ Password must contain at least one digit.
→ Password must contain at least one special character.

PASS | Test 2 - No Uppercase
→ Password must contain at least one uppercase letter.

PASS | Test 3 - No Lowercase
→ Password must contain at least one lowercase letter.

PASS | Test 4 - No Digit
→ Password must contain at least one digit.

PASS | Test 5 - No Special Char
→ Password must contain at least one special character.

PASS | Test 6 - Matches Username
→ Password must not match the username.

PASS | Test 7 - Valid Password

PASS | Test 8 - Null Password
→ Password must be at least 8 characters long.

Results
Passed: 8
Failed: 0
@Youssof-Hamed →/workspaces/maintenance (main) $
```

Validation:

- Original problem: Users could set weak passwords like 123 or abc with no restrictions
- Solution proves valid because: The validator now rejects any password that does not meet the 6 security rules
- Result: Weak passwords are no longer accepted by the system

⇒ Audit Logging Service

Verification: Unit Test Results

Test No.	Action	Expected log	Result
1	CREATE	Log written with timestamp&user name	<u>pass</u>
2	UPDATE	Log show old and new value	<u>pass</u>
3	DELETE	Log written with DELETE action	<u>pass</u>

The screenshot displays a VS Code editor interface with a terminal window open. The terminal shows the execution of unit tests for the Audit Logging Service. The tests are as follows:

```
Test 1 - Log CREATE:
[AUDIT] [2025-05-07 13:32:00] User=admin Action=Create Entity-Class ID=101 Old=[N/A] New=[CS101 Monday 9AM Room A] IP=192.168.1.1
PASS | CREATE Logged

Test 2 - Log UPDATE:
[AUDIT] [2025-05-07 13:32:00] User=manager1 Action=update Entity=Room ID=202 Old=[Room A Capacity 30] New=[Room B Capacity 50] IP=192.168.1.2
PASS | UPDATE Logged

Test 3 - Log DELETE:
[AUDIT] [2025-05-07 13:32:00] User=admin Action=Delete Entity-Class ID=303 Old=[CS202 Friday 2PM] New=[N/A] IP=192.168.1.3
PASS | DELETE Logged

Results Summary:
Passed: 3 | Failed: 0
Check timetable_audit.log for entries.
```

The terminal also shows the command to run the tests: `@bassel04 → /workspaces/maintenance (feature/audit-logging) $ java -cp out org.unimta.util timetable.util.AuditLogServiceTest`.

The Explorer panel on the left shows the project structure, including the `timetable_audit.log` file. The Chat panel on the right displays a "Build with Agent" message.

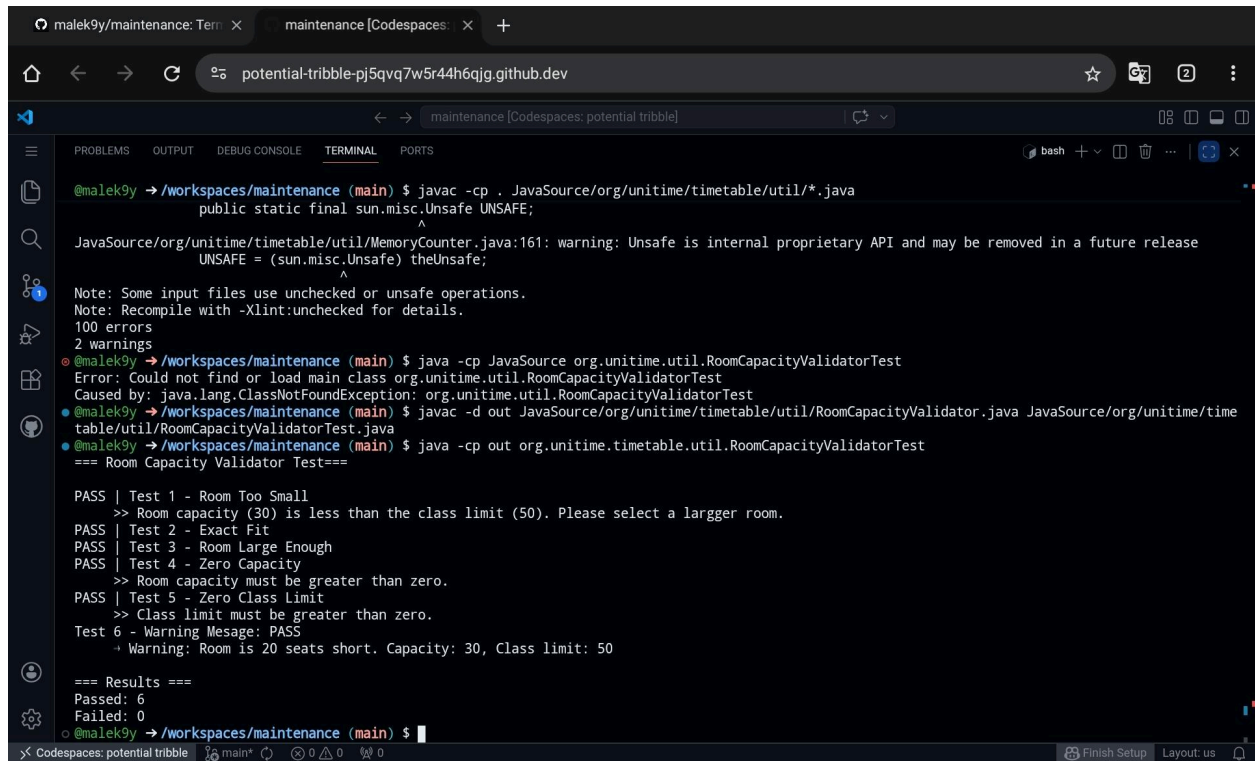
Validation:

- Original problem: No record of who changed what and when in the timetable
- Solution proves valid because: Every CREATE, UPDATE, and DELETE action is now recorded with the username, timestamp, and affected entity
- Result: A full audit trail now exists for all timetable changes

⇒Room Capacity Validation

Verification: Unit Test Results

Test No.	Test name	Room Capacity	Class limit	Expected	Result
1	Room too small	30	50	Invalid-Error message	<u>Pass</u>
2	Exact fit	50	50	Valid	<u>Pass</u>
3	Room large enough	100	50	Valid	<u>Pass</u>
4	Zero capacity	0	50	Invalid-Zero Not allowed	<u>Pass</u>
5	Zero class limit	50	0	Invalid-Zero Not allowed	<u>Pass</u>
6	Warning message	30	50	Warning: Room is 20 seats short	<u>Pass</u>



The screenshot shows a VS Code terminal window with the following content:

```
@malek9y → /workspaces/maintenance (main) $ javac -cp . JavaSource/org/unitime/timetable/util/*.java
public static final sun.misc.Unsafe UNSAFE;
                ^
JavaSource/org/unitime/timetable/util/MemoryCounter.java:161: warning: Unsafe is internal proprietary API and may be removed in a future release
    UNSAFE = (sun.misc.Unsafe) theUnsafe;
                ^
Note: Some input files use unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.
100 errors
2 warnings
@malek9y → /workspaces/maintenance (main) $ java -cp JavaSource org.unitime.util.RoomCapacityValidatorTest
Error: Could not find or load main class org.unitime.util.RoomCapacityValidatorTest
Caused by: java.lang.ClassNotFoundException: org.unitime.util.RoomCapacityValidatorTest
@malek9y → /workspaces/maintenance (main) $ javac -d out JavaSource/org/unitime/timetable/util/RoomCapacityValidator.java JavaSource/org/unitime/timetable/util/RoomCapacityValidatorTest.java
@malek9y → /workspaces/maintenance (main) $ java -cp out org.unitime.timetable.util.RoomCapacityValidatorTest
=== Room Capacity Validator Test===

PASS | Test 1 - Room Too Small
    >> Room capacity (30) is less than the class limit (50). Please select a largger room.
PASS | Test 2 - Exact Fit
PASS | Test 3 - Room Large Enough
PASS | Test 4 - Zero Capacity
    >> Room capacity must be greater than zero.
PASS | Test 5 - Zero Class Limit
    >> Class limit must be greater than zero.
Test 6 - Warning Mesage: PASS
    - Warning: Room is 20 seats short. Capacity: 30, Class limit: 50

=== Results ===
Passed: 6
Failed: 0
@malek9y → /workspaces/maintenance (main) $
```

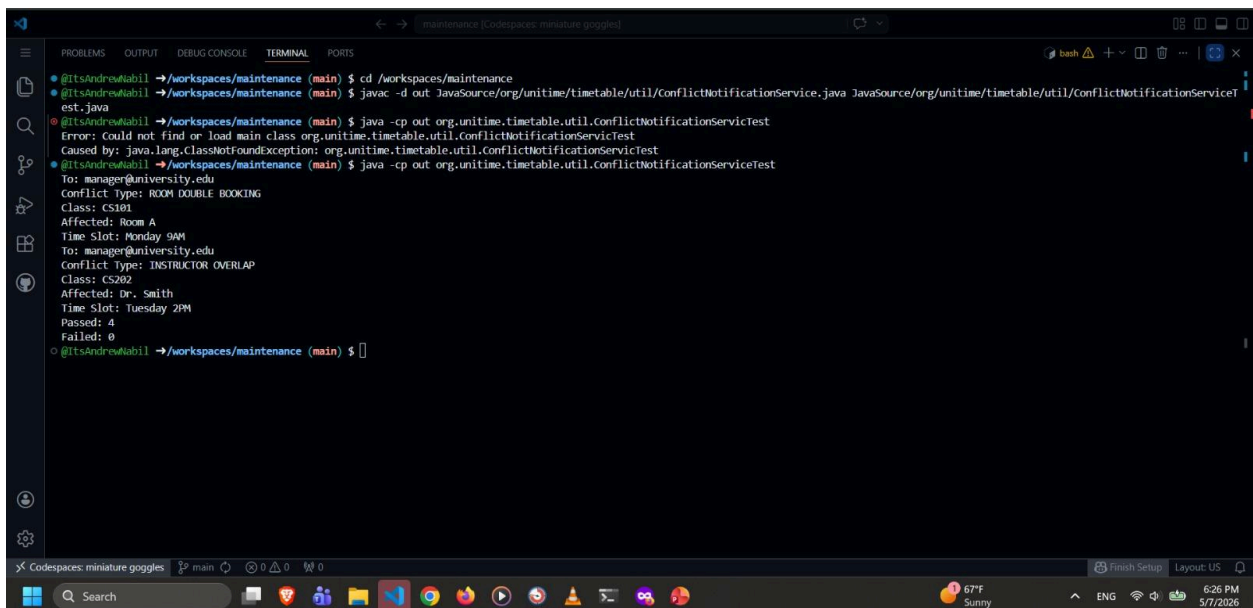
Validation:

- Original problem: A class with 50 students could be assigned to a room with capacity 30 and the system accepted it silently
- Solution proves valid because: The validator now catches this and returns a clear error message before the assignment is saved
- Result: Room assignments that cannot fit the class are now blocked with an informative message

⇒ Conflict Notification Service

Verification: Unit Test Results

Test No.	Conflict	Expected	Result
1	Double booking	Notification Generated with room details	<u>Pass</u>
2	Instructor overlap	Notification Generated with instructor details	<u>Pass</u>
3	Storage check	2 Notifications found after test 1,2	<u>Pass</u>
4	Clear Check	Notifications list emptied	<u>Pass</u>



```
maintenance [Codespaces: miniature goggles]
@ItsAndrewAbill →/workspaces/maintenance (main) $ cd /workspaces/maintenance
@ItsAndrewAbill →/workspaces/maintenance (main) $ javac -d out JavaSource/org/unitime/timetable/util/ConflictNotificationServiceTest.java
@ItsAndrewAbill →/workspaces/maintenance (main) $ java -cp out org.unitime.timetable.util.conflictNotificationServiceTest
Error: Could not find or load main class org.unitime.timetable.util.conflictNotificationServiceTest
Caused by: java.lang.ClassNotFoundException: org.unitime.timetable.util.conflictNotificationServiceTest
@ItsAndrewAbill →/workspaces/maintenance (main) $ java -cp out org.unitime.timetable.util.conflictNotificationServiceTest
To: manager@university.edu
Conflict Type: ROOM DOUBLE BOOKING
Class: CS101
Affected: Room A
Time Slot: Monday 9AM
To: manager@university.edu
Conflict Type: INSTRUCTOR OVERLAP
Class: CS202
Affected: Dr. Smith
Time Slot: Tuesday 2PM
Passed: 4
Failed: 0
@ItsAndrewAbill →/workspaces/maintenance (main) $
```

Validation

- Original problem: When a conflict was detected, only the UI highlighted it. Managers not actively using the system never found out
- Solution proves valid because: The service now generates a notification message containing the conflict type, affected class, room or instructor, time slot, and manager email
- Result: Managers are now immediately informed of scheduling conflicts

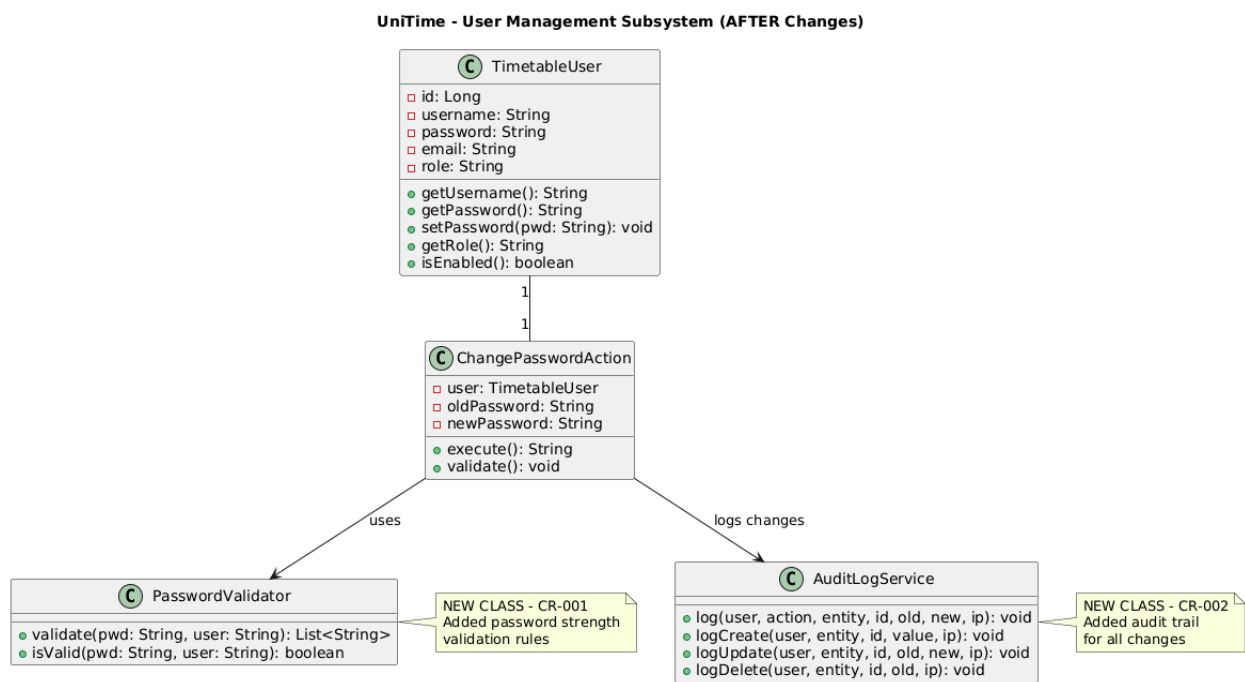
4-Github

Repository:

<https://github.com/malek9y/maintenance.git>

- Completed all assigned team issues successfully
- Implemented all 4 planned features/bug fixes
- Added and completed unit tests for all implemented features and fixes
- Verified functionality and test coverage for each implementation
- Ensured all changes were integrated and working as expected
- Closed related development tasks after successful implementation and testing

5-Uml Diagram after Changes



6-JIRA after Changes

<https://unitime-maintenance.atlassian.net/jira/software/projects/SCRUM/boards/1?atlOrigin=eyJpIjoiZDM3YjRlZmY2ZDFkNGU2YWUwMGYzMDU5Yjk5NWE5NGEiLCJwIjoiaaiJ9>

The screenshot displays the Jira interface for a SCRUM board. The browser address bar shows the URL: `unitime-maintenance.atlassian.net/jira/software/projects/SCRUM/boards/1`. The left sidebar contains navigation options: Jira, For you, Recent, Starred, Apps, Plans, Spaces, Recent, UniTime Maintenance, More spaces, Recommended, Collect requests, Filters, Dashboards, Teams, and Customize sidebar. The main content area shows the 'UniTime Maintenance' space with tabs for Summary, Backlog, Board, Code, Timeline, Docs, and Forms. The 'Board' tab is active, displaying a SCRUM board with columns: DONE (4 items), TO DO, IN PROGRESS, and IN REVIEW (2 items). The 'DONE' column contains four items:

- Item 1: Add password strength Validator for user password changes. Key: security-new-feature. Status: SCRUM-1. Assignee: BA.
- Item 2: No Validation when manually assigning a room smaller than class enrollment limit. Key: bugvalidationroom-mana... Status: SCRUM-2. Assignee: BA.
- Item 3: Send email notification to department manager when scheduling conflict is detected. Key: SCRUM-3. Assignee: BA.
- Item 4: Implement audit Logging for all time table change events. Key: loggingnew-featuredatab... Status: SCRUM-4. Assignee: BA.

The 'TO DO' column has a '+ Create' button. The 'IN PROGRESS' and 'IN REVIEW' columns are currently empty. The top right of the board shows a 'Complete sprint' button and a 'Group' dropdown menu.